Physics 9th – Chapter-4-Dynamics-II (Short Response Questions)

Q1. Why long spanner is used to open or tight nuts of vehicle's tyre? While tightening a small nut, extra-long wrench is not suitable. Why?

Ans:

- Moment of force $\tau = F \times d$; a long spanner has a large moment arm (d), so a large torque is produced with a small force.
- Tyre nuts are very tight, so a long spanner makes them easy to open or tighten.
- For a small nut, an extra-long wrench gives too much torque, which can over-tighten, break the nut or damage its threads; it is also hard to handle in small spaces.

Q2. Why door knobs are fixed at the edge of door? What will happen if the door knob is at the middle of the door?

Ans:

- The axis of rotation of a door is at its hinges; a knob at the far edge gives the maximum moment arm.
- So the door opens with the least force ($\tau = F \times d$).
- If the knob is in the middle, the moment arm is halved, so about double force is needed to open the door.

Q3. If you drop a feather and a bowling ball from the same height, which one will reach the terminal velocity first? Which one of them will hit the ground first?

Ans:

- The feather reaches terminal velocity first, because its small weight is soon balanced by air resistance (drag), so its terminal velocity is very low.
- The bowling ball hits the ground first; its large weight is much greater than air drag, so it keeps accelerating.
- Concept: in vacuum (no air resistance), both would fall together with acceleration g.

Q4. Why do ice skates effortlessly slide on ice, while your shoes cause skidding?

Ans:

- The thin blades of skates exert high pressure, melting a thin layer of ice; this water layer acts as a lubricant and reduces friction, so skates glide smoothly.
- Shoe soles have a large area and get very little grip on smooth ice, so they slip in an uncontrolled way (skid).
- Concept: friction depends on the nature of surfaces in contact; lubricants reduce friction.

Q5. Explain why it's easier to push a car on flat tyres than inflated ones. What happens to the frictional force between the tyres and the road?

Ans:

- In fact, it is harder to push a car with flat tyres.
- Flat tyres deform (flatten) more at the contact with the road, which increases rolling friction (rolling resistance).
- Properly inflated tyres deform less, so rolling friction is smaller and the car is easier to push.

- Concept: rolling friction is much less than sliding friction, but it increases with deformation of the tyre.

Q6. When standing on a crowded school bus, which stance would provide better stability and prevent you from being pushed over: legs joined or legs spread apart?

Ans:

- Legs spread apart give better stability.
- It increases the area (base) of support and slightly lowers the centre of gravity.
- The line of action of weight stays within the base for a larger tilt, so you are not easily toppled.

Q7. Why a moving bicycle is easier to balance? Relate this to the principles of rotational motion.

Ans:

- The spinning wheels of a moving bicycle tend to keep rotating about the same axis (rotational inertia, the rotational form of Newton's first law).
- This resists tilting (gyroscopic effect), and the rider can also steer to keep the centre of gravity above the wheels.
- A bicycle at rest has no such rotational effect, so it falls easily.

Q8. Why is a pencil standing on its tip unstable, and what factors affect the stability of an object balanced on a point?

Ans:

- The base (tip) is very small and the centre of gravity is high, so a slight tilt brings the CG outside the base.
- Its weight then produces a moment that increases the displacement, so the pencil falls: unstable equilibrium (CG falls when displaced).
- Factors: height of the centre of gravity, area of the base, and position of the CG relative to the point of support.

Q9. While driving what happens if the driver takes the curve too fast? How does centripetal force play a role in keeping the car from skidding off the road?

Ans:

- On a curve, the centripetal force ($F = mv^2/r$) is provided by friction between the tyres and the road.
- At high speed the required centripetal force becomes larger than the maximum friction available.
- The car then cannot follow the curve, moves outward along a straight line due to inertia and skids off the road; so drivers slow down on turns.

Q10. Consider a situation where you swing a ball connected to a string in a circle. How does the tension in the string vary as the ball moves across different points in its circular path, and what forces are involved?

Ans:

- Forces involved: tension in the string and the weight of the ball; their resultant towards the centre gives the centripetal force (mv^2/r).
- In a vertical circle, tension is maximum at the lowest point ($T = mv^2/r + mg$) and minimum at the highest point ($T = mv^2/r - mg$).
- In a horizontal circle at constant speed, the tension remains almost constant.

Q11. Why is it important for communication satellites in geostationary orbit to maintain a specific speed?

Ans:

- A geostationary satellite must complete one orbit in 24 hours, the same as Earth's rotation, so it stays above the same point on Earth.
- This needs a fixed orbital speed (about 3.1 km/s at about 36 000 km height), where gravity provides exactly the required centripetal force.
- If its speed is more, it moves to a higher/elliptical orbit or escapes; if less, it falls to a lower orbit, and the communication link is lost.